

Project Design Phase – II

Solution Requirements (Functional & Non-Functional)

Date: 25 June 2026

Team ID: XXXXX

Project Name: Smart Lender

Maximum Marks: 4 Marks

Functional Requirements

The following are the functional requirements of the **Smart Lender** system.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Home Page	Display project overview and provide a button to navigate to the loan prediction page.
FR-2	Loan Application Form	Allow users to enter applicant details such as Gender, Married Status, Dependents, Education, Self-Employment, Income, Loan Amount, Credit History, and Property Area.
FR-3	Input Validation	Verify that all mandatory fields are completed and that numeric values are valid before processing.
FR-4	Loan Prediction	Send the applicant's data to the trained Random Forest (rdf.pkl) model and generate a prediction.
FR-5	Prediction Result	Display whether the loan is Approved or Rejected , along with a clear message to the user.
FR-6	Error Handling	Display appropriate error messages for invalid or incomplete input and prevent incorrect submissions.

Non-Functional Requirements

The following are the non-functional requirements of the proposed solution.

NFR No.	Requirement	Description
NFR-1	Usability	The application should have a clean, simple, and user-friendly interface that is easy to navigate.
NFR-2	Performance	Loan prediction results should be generated within a few seconds after the user submits the form.
NFR-3	Accuracy	The Random Forest model should provide accurate and consistent loan approval predictions based on the trained dataset.
NFR-4	Reliability	The system should operate consistently without crashes during normal usage.
NFR-5	Security	User input should be handled securely, and sensitive information should not be exposed.
NFR-6	Scalability	The application should support multiple users and allow future enhancements such as user authentication, database integration, and cloud deployment.